



Antidiabetic effect of *Ficus bengalensis* aerial roots in experimental animals

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ABSTRACT

Ethnopharmacological relevance: Herbal preparations of *Ficus bengalensis* had been considered as effective, economical and safe ethnomedicines for various ailments in Indian traditional system of medicine.

Aim of study: The present study was aimed to explore scientifically the antidiabetic potential of *Ficus bengalensis* aerial roots as its bark had already been reported to possess antidiabetic efficacy.

Materials and methods: Effect of variable doses of aqueous extract of *Ficus bengalensis* aerial roots on blood glucose level (BGL) of normal-, sub- and mild-diabetic models have been studied and the results were compared with the reference drug Glipizide and elemental Mg and Ca intake as glycemic elements.

Results: The dose of 300 mg kg⁻¹ showed the maximum fall of 43.8 and 40.7% in BGL during FBG and glucose tolerance test (GTT) studies of normal rats, respectively. The same dose showed a marked reduction in BGL of 54.3% in sub- and 51.7% in mild-diabetic rats during GTT. The concentration of Mg (1.02%) and Ca (0.85%) identified through laser induced breakdown spectroscopy (LIBS) in the most effective dose could be responsible for this high percentage fall in BGL as they take part in glucose metabolism.

Conclusion: The hypoglycemic effect in normoglycemic and antidiabetic effect in sub- and mild-diabetic models of aqueous extract of aerial roots of *Ficus bengalensis* are due to the presence of these glycemic elements in high concentration with respect to other elements.

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1. Introduction

Diabetes mellitus is a global burden as its incidence is considered to be high (4–5%) all over the world (Pickup and William, 1997). However, quest for the development of more effective antidiabetic agents is being pursued relentlessly (Ghosh et al., 2004). Recently, herbal products have started gaining importance as complementary and alternative medicine to treat diabetic mellitus (Payne, 2001; Rai et al., 2007a). Though, many herbal products have been described for the treatment of diabetic mellitus, very few of them have been explored scientifically so far. Biological activities of medicinal plants are closely related to their elemental composition. Plants rich in Mg and Ca generally have high potential of lowering blood glucose level (BGL) (Rai et al., 2007b).

Ficus bengalensis Linn. Family: (Moraceae) is a very large tree distributed throughout India. It is commonly known as 'Bargad' in Hindi or 'Indian Banyan tree' and considered as holy tree of India. Information based on ethnomedicinal survey reveals that the herbal preparations of different parts of *Ficus bengalensis* had been consid-

ered as effective economical and safe treatments for curing various diseases in Indian traditional system of medicine.

The hanging roots of *Ficus bengalensis* have been reported as anti-diarrhoeal agents (Mukherjee et al., 1998). The fruit extract of *Ficus bengalensis* has been documented for its anti-tumor and anti-bacterial activities (Mousa et al., 1994). The plant is used in folk medicine for respiratory disorders and certain skin diseases (Kirtikar and Basu, 1935). Bark of *Ficus bengalensis* has been traditionally used for the management of diabetes mellitus. Oral administration of bark extract showed lowering of blood glucose level in STZ diabetic animals and enhancement of serum insulin levels in normoglycemic as well as diabetic rats (Achrekar et al., 1991). Blood sugar lowering and serum insulin raising action was also found in a dimethoxy derivative of leucocyanidin 3-O-beta-D-galactosyl cellobioside (Kumar and Augusti, 1989) and a dimethoxy ether of leucopelargonidin-3-O-alpha-L-rhamnoside isolated from the bark of *Ficus bengalensis*. Bengalenoside, a glucoside isolated from *Ficus bengalensis* also showed hypoglycemic activity in normal and alloxan diabetic rabbits (Augusti, 1975). Anti-oxidant effect of aqueous extract of the bark of *Ficus bengalensis* has been evaluated in hypercholesterolemic rabbits (Shukla et al., 2004).

Since, no work has been carried out so far on aerial roots of *Ficus bengalensis* for diabetes management therefore the present study was undertaken to evaluate the glycemic profile of the aqueous extract of *Ficus bengalensis* aerial roots on blood glucose level of normoglycemic and streptozotocin (STZ) induced hyperglycemic

Abbreviations: FBG, fasting blood glucose; BGL, blood glucose level; GTT, glucose tolerance test; STZ, streptozotocin; LD₅₀, lethal dose₅₀; LIBS, laser induced breakdown spectroscopy.

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rats. The study is based on their glucose tolerance test (GTT) studies along with the glycemic element identification by laser induced breakdown spectroscopy (LIBS) technique. Phytochemical nature of the aerial roots extract has also been carried out based on qualitative chemical tests.

2. Materials and methods

2.1. Plant material

Fresh aerial roots of *Ficus bengalensis* were collected and identified by Prof. Satya Narayan, Taxonomist, Department of Botany, University of Allahabad, Allahabad, India. The roots were dried and cut into small pieces, the pieces were mechanically crushed. 4 kg of crushed aerial roots were continuously extracted with distilled water using soxhlet up to 48 h. The extract was filtered and concentrated in rotatory evaporator at 35–40 °C under reduced pressure to obtain a semisolid material, which was then lyophilized to get a powder (12.32%, w/w).

2.2. Chemicals

1,1-Diphenyl-2-picryl hydrazyl (DPPH) and quercetin were purchased from Sigma Chemical Co. (St., Louis, USA). Gallic acid, tert-butyl-4-hydroxy toluene (BHT), Folin Ciocalteu reagent, and methanol were purchased from Merck Co. (Germany).

2.3. Experimental animals

Female albino Wistar rats of approximately same age group, having body weight 210–250 g were obtained from National Institute of Communicable Disease (NICD) Delhi, and were used in the experiment. Animals were kept in our animal house at an ambient temperature of $27 \pm 3^\circ\text{C}$ and $50 \pm 5\%$ relative humidity with a 12 h each of dark and light cycle. Animals were fed with pellet diet (Pashu Aahar Kendra, Varanasi) and distilled water. The study was approved by the Institutional Ethical Committee.

2.4. Induction of diabetes in rats

Diabetes was induced by single intraperitoneal injection of freshly prepared solution of STZ at the dose of 45 mg kg^{-1} in 0.1 M citrate buffer (pH 4.5) to the rats fasted overnight. After 3 days of STZ induction, FBG was checked and animals were divided into two groups, sub-diabetic and mild-diabetic. Animals showing normal FBG but abnormal GTT were considered as sub-diabetic and animals with abnormal FBG ($110\text{--}250 \text{ mg dl}^{-1}$) and abnormal GTT were classified as mild-diabetic rats (Kesari et al., 2005).

2.5. Estimation of BGL and detection of trace elements

Blood glucose level was estimated by glucose oxidase method (Barham and Trinder, 1972) using standard kit of Bayer Diagnostics India Limited, New Delhi, India. Trace elements were detected by laser induced breakdown spectroscopy using Ocean optics LIBS 2000+ equipped with CCD.

2.6. Experimental design

Screening of the extract for hypoglycemic activity was done with a range of variable doses ($100, 200, 300$ and 400 mg kg^{-1}) in normal healthy rats by conducting fasting blood glucose (FBG) and glucose tolerance test studies. Whereas, the antidiabetic action of the extract was assessed in sub- and mild-diabetic models by conducting GTT studies with the same range of doses. Two groups of rats

were administered with elemental Mg and Ca separately for each models: normal-, sub- and mild-diabetic.

2.6.1. Assessment of hypoglycemic potential in normal rats

Five groups of six rats each were fasted overnight. Group 1 served as untreated control and received vehicle (distilled water) only. Animals of group 2, 3, 4 and 5 received graded doses of $100, 200, 300$ and 400 mg kg^{-1} respectively of aqueous aerial root extract powder suspended in distilled water. Blood samples were collected from tail vein at 0 h and then at 2, 4, 6 and 8 h after extract administration for FBG studies.

For GTT studies aqueous extract was given orally to different groups of healthy animals in the same fashion as above and their effect on FBG was studied hourly up to 2 h. The BGL value at 2 h was treated as 0 h value for GTT. The animals were orally administered with 4 g kg^{-1} of glucose and their glucose tolerance was studied at 1 h intervals for another 3 h. Thus, total period of blood collection was up to 5 h.

2.6.2. Assessment of antidiabetic potential in sub- and mild-diabetic rats

The antidiabetic effect of aqueous extract in sub- and mild-diabetic rats was assessed by improvement in glucose tolerance. The rats of both sub- and mild-diabetic models were divided into six groups of six rats each. Group 1 is control, received vehicle (distilled water only). Whereas variable doses of $100, 200, 300, 400 \text{ mg kg}^{-1}$ of aerial root extract was given orally to group 2, 3, 4 and 5, respectively. Blood glucose levels were checked firstly after 90 min of treatment considered as 0 h value and then 2 g kg^{-1} glucose was given orally to all the groups. Blood glucose levels were further checked up to 3 h at regular intervals of 1 h each, considered as 1, 2 and 3 h values. The results were compared with six groups of rats, treated with 2.5 mg kg^{-1} of Glipizide, a reference drug.

2.6.3. Assessment of glycemic elements potential

Three groups of six rats each for each models; normal- (group 1, 2, 3) sub- (4, 5, 6) and mild-diabetic (7, 8, 9) were fasted overnight. Group 1, 4 and 7 served as control for normal-, sub- and mild-diabetic animals, respectively. Whereas, group 2, 5 and 8 received Mg at a dose of 1.02 mg kg^{-1} and group 3, 6 and 9 received Ca at a dose of 0.85 mg kg^{-1} .

2.6.4. Preliminary phytochemical investigation

Phytochemical analysis of the crude extract for phenolic and flavonoids was determined according to Kokte (1994) and Harborne (1998). The plant extract powder (200 mg) was dissolved in 100 ml ethanol and filtered. 2 ml of this filtrate was mixed with equal volume of concentrated HCl followed by addition of the magnesium ribbon. The appearance of tomato red color indicated the presence of flavonoids in the extract of *Ficus bengalensis*. The phenolic extraction of the dried powder sample has been performed using 70% ethanol and total phenolic content was analyzed, using Folin Ciocalteu reagent (Mc Donald et al., 2001).

2.6.5. Statistical analysis

Statistical analysis was performed using two-way analysis of variance (ANOVA), using statistical package PRISM 3.0 version. The significance of difference between and within various groups was determined. Differences were considered to be significant when $P < 0.05$.

2.6.6. LD₅₀ experiment

Two groups of six rats each of both the sex were orally administered with a single dose of 10 and 15 times of the most effective dose of aqueous extract of *Ficus bengalensis* aerial roots. The rats

Table 1Effect of graded doses of aqueous extract of *Ficus bengalensis* aerial roots on BGL during FBG and GTT of normoglycemic rats (mean $\pm$ S.D.).

Exp group	Treatment	BGL of normoglycemic (mg dl ⁻¹) during FBG				
		FBG	2 h	4 h	6 h	8 h
Group 1	Control	78.7 $\pm$ 4.5	78.3 $\pm$ 3.2	78.9 $\pm$ 4.1	79.0 $\pm$ 2.7	78.1 $\pm$ 4.1
Group 2	100	76.5 $\pm$ 3.9	73.2 $\pm$ 2.9	63.8 $\pm$ 3.4	52.1 $\pm$ 2.8*	58.5 $\pm$ 4.3*
Group 3	200	77.3 $\pm$ 3.8	72.5 $\pm$ 3.6	60.2 $\pm$ 4.2*	48.2 $\pm$ 3.4	56.8 $\pm$ 3.5
Group 4	300	76.9 $\pm$ 3.1	70.6 $\pm$ 3.3	52.7 $\pm$ 3.6*	43.2 $\pm$ 3.5*	55.7 $\pm$ 3.7**
Group 5	400	75.6 $\pm$ 4.7	69.8 $\pm$ 3.5	51.8 $\pm$ 3.8	44.4 $\pm$ 3.7	56.5 $\pm$ 3.8*

Exp group	Treatment	BGL of normoglycemic animals (mg dl ⁻¹) during GTT				
		FBG	0 h	1 h	2 h	3 h
Group 1	Control	75.7 $\pm$ 3.2	75.4 $\pm$ 2.7	108.2 $\pm$ 2.5	74.5 $\pm$ 3.6	78.5 $\pm$ 3.5
Group 2	100	77.3 $\pm$ 2.9	74.0 $\pm$ 3.1	94.5 $\pm$ 4.1	68.7 $\pm$ 2.5	50.2 $\pm$ 2.7
Group 3	200	79.6 $\pm$ 3.6	73.6 $\pm$ 3.3	90.5 $\pm$ 4.1	63.2 $\pm$ 3.8**	59.4 $\pm$ 3.2*
Group 4	300	75.4 $\pm$ 3.8	69.2 $\pm$ 3.4	78.6 $\pm$ 3.2	54.9 $\pm$ 3.8*	46.5 $\pm$ 3.7**
Group 5	400	76.5 $\pm$ 4.6	70.7 $\pm$ 2.9	80.4 $\pm$ 3.9	58.1 $\pm$ 4.1*	48.2 $\pm$ 4.2**

The signs (**) and (*) indicate values significantly different from initial and control at $P < 0.01$ and $P < 0.05$ during FBG and GTT.

were observed for their gross behavioral, neurologic, autonomic and toxic effects at short intervals of time up to 48 h. Food consumption, faeces and urine were also examined at 2 h and then 6 h interval for 48 h.

3. Results

3.1. Effect on FBG of normal healthy rats

Table 1 shows the effect of graded doses of aqueous extract of *Ficus bengalensis* aerial roots on FBG level of normal healthy rats. All the four doses of 100, 200, 300 and 400 mg kg⁻¹ produced significant fall at 6 h of oral administration.

The dose of 300 mg kg⁻¹ showed the maximum fall of 43.8% whereas a fall of 31.8, 37.6 and 41.2% was observed with the doses of 100, 200 and 400 mg kg⁻¹ at 6 h of oral administration. However, rise in BGL was observed after 6 h of extract administration.

3.2. Effect on glucose tolerance of normal healthy rats

Table 1 depicts the result of GTT studies of normal healthy rats. The maximum fall of 40.7% was observed with the dose of 300 mg kg⁻¹ whereas, doses of 100, 200 and 400 mg kg⁻¹ produced a fall of 14.5, 23.1 and 38.5% respectively after 3 h of glucose administration. This study also supports that 300 mg kg⁻¹ is the most effective dose.

3.3. Effect on glucose tolerance of sub- and mild-diabetic rats

Figs. 1 and 2 reveals the effect of graded doses of aqueous extract of *Ficus bengalensis* on BGL of sub- and mild-diabetic animals during GTT. Different doses of aqueous extract of 100, 200, 300 and 400 mg kg⁻¹ and an synthetic drug, Glipizide (2.5 mg kg⁻¹) were given orally to the different groups. The fall of 31.1, 43.8, 54.3 and 53.9% in BGL of sub-diabetic rats was observed after 3 h of glucose administration with the doses of 100, 200, 300 and 400 mg kg⁻¹, respectively. The dose of 2.5 mg kg⁻¹ of Glipizide reduced BGL by 33.8% at 3 h during GTT in sub-diabetic rats. The fall observed in BGL of mild-diabetic rats after glucose administration was 41.8, 46.4, 51.7 and 49.8% with the dose of 100, 200, 300 and 400 mg kg⁻¹, respectively. This confirms that the dose of 300 mg kg⁻¹ of aqueous extract is the most effective dose. Moreover, the dose of 2.5 mg kg⁻¹ of Glipizide produced a fall of 49.1% in mild-diabetic rats during GTT. The fall produced in BGL by the dose of 300 mg kg⁻¹ is

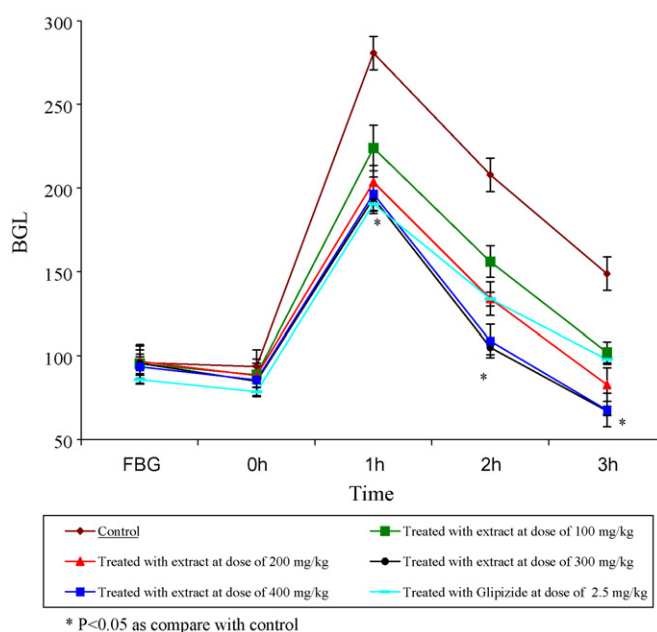


Fig. 1. Effect of graded doses of aqueous extract of *Ficus bengalensis* aerial roots on BGL during GTT of sub-diabetic rats.

higher as compared to that of standard drug Glipizide in case of sub-diabetic animals. However, in case of mild-diabetic rats the effect of 300 mg kg⁻¹ was almost similar to the standard drug Glipizide.

3.4. Detection of mineral elements

The LIBS results showed much higher concentration of Mg and Ca in aqueous extract of *Ficus bengalensis* than the other elements present.

3.5. Effect of glycemic elements on BGL of normal-, sub- and mild-diabetic models

Table 2 reveals the effect of oral administration of Mg and Ca in normal-, sub- and mild-diabetic models during FBG and GTT studies. The maximum fall of 6.2, 12.5 and 16.8% was observed in normal-, sub- and mild-diabetic rats respectively with the dose of 1.02 mg kg⁻¹ of elemental Mg and a fall of 4.6, 14.4 and 18.2% was

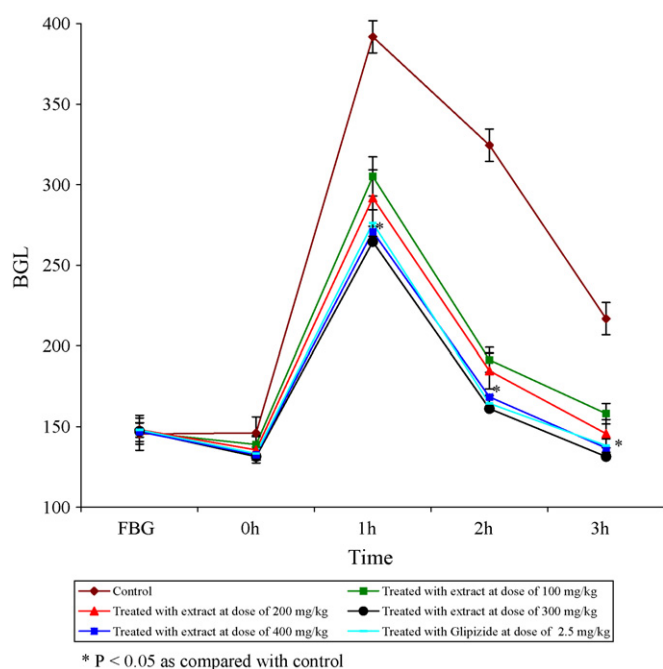


Fig. 2. Effect of graded doses of aqueous extract of *Ficus bengalensis* aerial roots on BGL during GTT of mild-diabetic rats.

observed in normal-, sub- and mild-diabetic rats respectively with the dose of 0.85 mg kg^{-1} of elemental Ca.

3.6. Phytochemical studies

The studies indicate that the total phenolic content in terms of gallic acid equivalent (mg g^{-1} of dry mass), is 25 mg g^{-1} in the extract powder.

3.7. LD₅₀

Experiment was carried out on normal healthy rats. The behavior of treated rats appeared normal. No toxic effect was observed at doses up to 10 and 15 times of effective dose of the aqueous extract. There was no death in any of these groups.

4. Discussion

The present investigation is the first reporting of antidiabetic action of aerial roots of *Ficus bengalensis* in normal and diabetic models. The results indicate that the extract of *Ficus bengalensis* decreases the blood glucose level in normal animals since the untreated control group showed higher BGL as compared to the treated groups. The maximum hypoglycemic activity in case of normal rats was observed by reducing BGL 43.8% at 6 h during FBG studies with the dose of 300 mg kg^{-1} . The effect was dose dependent up to 300 mg kg^{-1} . However, the response decreases at higher dose of 400 mg kg^{-1} . Such a phenomenon of less hypoglycemic response at higher doses is not uncommon with indigenous plants and has already been observed in *Murraya koenigii* (Kesari et al., 2005), *Cynodon dactylon* (Singh et al., 2007), *Trichosanthes dioica* (Rai et al., 2008) and *Aegle marmelos* (Kesari et al., 2006). The dose of 300 mg kg^{-1} also showed a marked improvement of 40.7, 54.8 and 51.7% in glucose tolerance of normal-, sub- and mild-diabetic animals at 3 h during GTT. Thus, the dose of 300 mg kg^{-1} was identified as the most effective dose in both, FBG as well as GTT studies. Moreover, fall produced in BGL by the most effective dose, was higher than the standard drug Glipizide (2.5 mg kg^{-1}) in the case of sub-diabetic rats and was almost same in mild-diabetic rats.

In normal physiology glucose homeostasis is maintained by two kinds of hormones, including insulin and counter-regulatory hormones (glucagons, growth hormone, cortisol and catecholamines) (Cryer and Polonsky, 1998; Gerich, 1988). Despite the presence of such counter-regulatory hormones, extract of aerial roots of *Ficus bengalensis* produced hypoglycemia, indicating thereby that the extract possess pharmacological activity, based on the suppression of gluconeogenesis (Pilkis et al., 1988; Kumar and Augusti, 1994). However, after 6 h of the extract administration the blood glucose level started increasing, which indicates that the counter-regulatory hormones overcome the hypoglycemia produced by the extract. The results of this study, conclusively reveal that the aqueous extract of *Ficus bengalensis* aerial roots have beneficial effect on lowering BGL.

These results were further confirmed by the LIBS results. Since, according to Boltzmann distribution law, intensity is directly related to concentration (Sabsabi and Cielo, 1995). Therefore, the concentration of major elements present in the extract can define their role in diabetes management. Hence, The high intensity peak of Mg and Ca clearly indicate the high concentration of both the elements with the respect to other elements in the extract,

Table 2
Effect of Mg and Ca on BGL during FBG and GTT of normal-, sub- and mild-diabetic rats (mean $\pm$ S.D.).

Exp group	Treatment	BGL of normoglycemic (mg dl^{-1}) during FBG				
		FBG	2 h	4 h	6 h	8 h
Group 1	Control	73.6 $\pm$ 3.2	74.7 $\pm$ 4.3	72.3 $\pm$ 3.6	73.9 $\pm$ 3.7	73.2 $\pm$ 3.5
Group 2	Mg	74.5 $\pm$ 3.6	74.1 $\pm$ 2.9	73.4 $\pm$ 3.9	72.0 $\pm$ 3.1*	71.3 $\pm$ 4.3*
Group 3	Ca	75.2 $\pm$ 3.5	74.4 $\pm$ 3.7	73.2 $\pm$ 3.3*	71.8 $\pm$ 4.2*	70.4 $\pm$ 3.0
Exp group	Treatment	BGL of sub-diabetic animals (mg dl^{-1}) during GTT				
		FBG	0 h	1 h	2 h	3 h
Group 4	Control	75.9 $\pm$ 4.2	74.5 $\pm$ 3.7	260.3 $\pm$ 3.8	222.4 $\pm$ 4.3	156.3 $\pm$ 4.1
Group 5	Mg	76.2 $\pm$ 3.2	73.9 $\pm$ 4.8	235.9 $\pm$ 5.5	202.4 $\pm$ 4.6*	136.8 $\pm$ 5.4
Group 6	Ca	75.4 $\pm$ 3.5	74.7 $\pm$ 4.3	232.7 $\pm$ 3.8	200.8 $\pm$ 5.2*	135.2 $\pm$ 3.9**
Exp group	Treatment	BGL of mild-diabetic animals (mg dl^{-1}) during GTT				
		FBG	0 h	1 h	2 h	3 h
Group 7	Control	152.36 $\pm$ 4.3	150.0 $\pm$ 3.5	380.4 $\pm$ 4.9	345.9 $\pm$ 5.2	235.7 $\pm$ 4.8
Group 8	Mg	154.4 $\pm$ 4.8	152.2 $\pm$ 5.1	365.4 $\pm$ 4.9	304.3 $\pm$ 5.3*	198.6 $\pm$ 6.2*
Group 9	Ca	150.9 $\pm$ 4.2	148.6 $\pm$ 5.2	361.8 $\pm$ 3.6	302.6 $\pm$ 4.7*	192.8 $\pm$ 3.3**

The signs (**) and (*) indicate values significantly different from initial and control at $P < 0.01$ and $P < 0.05$ during FBG and GTT.

responsible for its antidiabetic potential, as Ca^{2+} activate the insulin gene expression via Calcium Responsive Element Binding Protein (CREB) resulting in exocytosis of stored insulin (Giugliano et al., 2000). Presence of Mg has also been correlated with the diabetes management in our earlier study (Rai et al., 2007a). The concentration of Mg and Ca in the most effective dose of 300 mg kg^{-1} was found to be 1.02 and 0.85% by calibration free (Ciucci et al., 1999) LIBS technique. It is interesting to note that the BGL falls when the same doses of elemental Mg and Ca were given to two different groups of animals. Further pharmacological and biochemical studies are in progress to elucidate the mechanism of action of the extract in detail at molecular level.

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